SALES DATA VISUALIZATION PROJECT

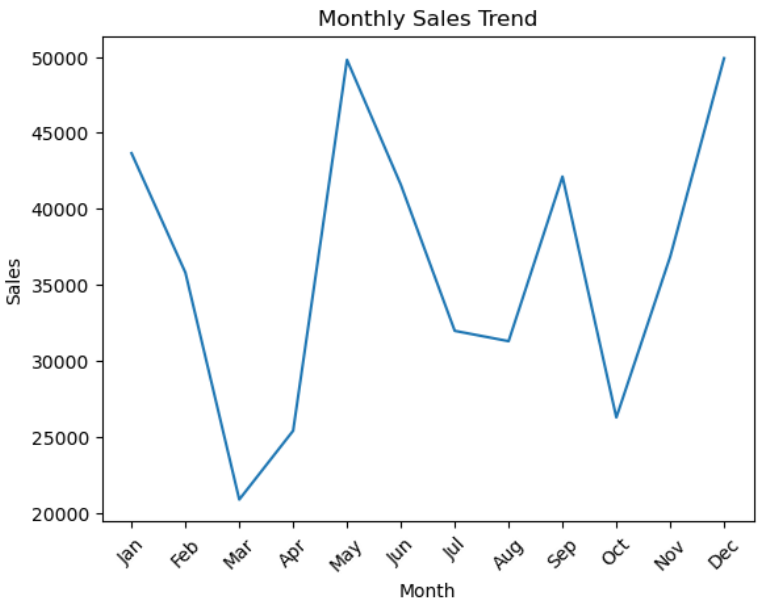
```
# import Libraries
import pandas as pd
import matplotlib.pyplot as plt

#  Load Dataset
df = pd.read_csv("sales_dataset.csv")

#  Display First Rows
print("First 5 Rows of Dataset:")
print(df.head())

# Basic Information
print("\nDataset Info:")
print(df.info())

print("\nStatistical Summary:")
print(df.describe())

#  LINE CHART – SALES TREND
plt.figure()
plt.plot(df["Month"], df["Sales"])
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.show()
```

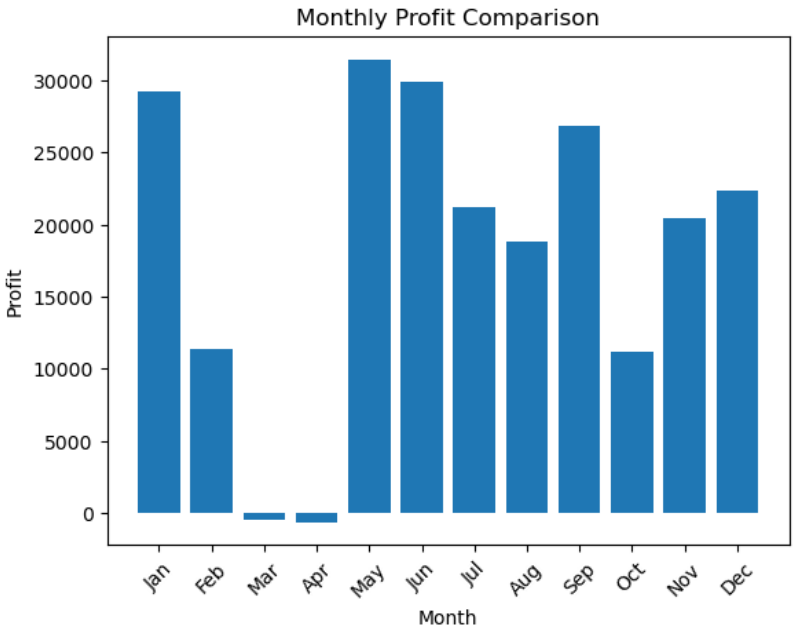
```
#  BAR CHART – PROFIT COMPARISON
plt.figure()
plt.bar(df["Month"], df["Profit"])
plt.title("Monthly Profit Comparison")
plt.xlabel("Month")
plt.ylabel("Profit")
plt.xticks(rotation=45)
plt.show()

#  SCATTER PLOT – SALES vs PROFIT
plt.figure()
plt.scatter(df["Sales"], df["Profit"])
plt.title("Sales vs Profit Relationship")
plt.xlabel("Sales")
plt.ylabel("Profit")
plt.show()

#  HISTOGRAM – SALES DISTRIBUTION
plt.figure()
plt.hist(df["Sales"], bins=8)
plt.title("Sales Distribution")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()
```

```
# CORRELATION ANALYSIS

correlation = df[["Sales", "Profit", "Expenses"]].corr()

print("\nCorrelation Matrix:")

print(correlation)
```

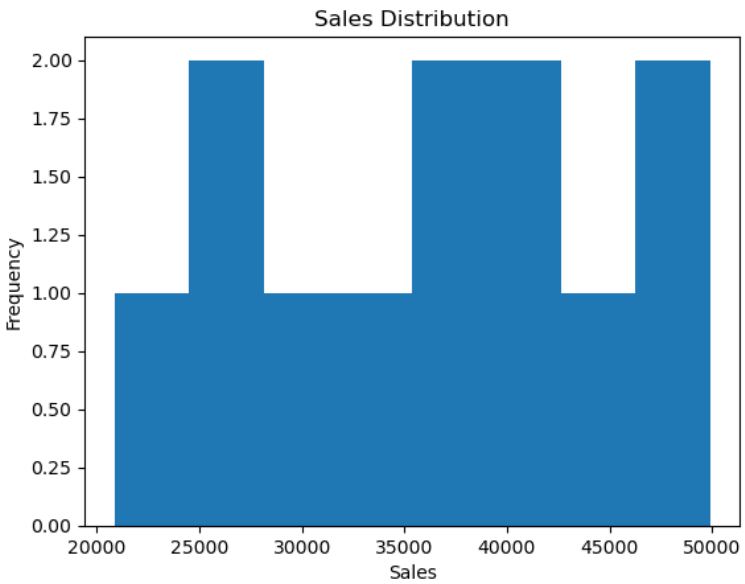
OUTPUT:

```
First 5 Rows of Dataset:
  Month  Sales  Expenses  Profit
0   Jan  43654     14426   29228
1   Feb  35795     24423   11372
2   Mar  20860     21363    -503
3   Apr  25390     26023    -633
4   May  49802     18322   31480

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12 entries, 0 to 11
Data columns (total 4 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Month     12 non-null     object
 1   Sales     12 non-null     int64
 2   Expenses  12 non-null     int64
 3   Profit    12 non-null     int64
dtypes: int64(3), object(1)
memory usage: 516.0+ bytes
None

Statistical Summary:
              Sales      Expenses       Profit
count     12.000000     12.000000     12.00000
mean   36288.916667  17816.166667  18472.75000
std     9467.323213   5740.868414  11034.04562
min    20860.000000  10769.000000   -633.00000
25%    30029.250000  13927.750000  11332.50000
50%    36322.500000  15865.500000  20812.50000
75%    42502.000000  22128.000000  27412.25000
max    49910.000000  27568.000000  31480.00000
```


Monthly Sales Trend
Sales
50000
45000
40000
35000
30000
25000
20000
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month


Monthly Profit Comparison
Profit
30000
25000
20000
15000
10000
5000
0
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month

Sales vs Profit Relationship
Profit
30000
25000
20000
15000
10000
5000
0
20000
25000
30000
35000
40000
45000
50000
Sales




```
Correlation Matrix:
             Sales    Profit  Expenses
Sales     1.000000  0.854001  0.007706
Profit    0.854001  1.000000 -0.513675
Expenses  0.007706 -0.513675  1.000000
```